

Real-Time Game Chat System – Short Summary

Goal: Provide a real-time, low-latency, secure chat system for games supporting party, squad, guild, match, global, and direct messages with presence, typing, and moderation. Must scale to millions of players across regions.

Key Outcomes: Sub-200ms latency, ordered delivery, exactly-once semantics, elastic fanout, strong moderation, operational excellence.

Assumptions: 1–10M concurrent players, 100K–1M msgs/sec peak, small + large rooms, multi-region deployment.

Functional Requirements: Messaging across channels, presence, typing, moderation (filters, mutes, bans, reports), persistence, offline delivery, search, announcements, game integration.

Non-Functional Requirements: 99.95% availability, p95 <100ms, strong ordering, zero data loss, multi-AZ durability, linear scalability, cost efficiency.

Architecture: Regional cells with gateways, messaging, presence, moderation, Redis caches, durable DB (DynamoDB/Scylla).
Global control plane for auth, config, routing. Clients via geo DNS/anycast.

Data Flow: Client → Gateway → Kafka/Pulsar → Fanout → Redis & DB → Pub/Sub → Websocket push or queued delivery.

Storage: Redis (hot history, presence), DynamoDB/Scylla (durable history), OpenSearch (search), Object storage (cold).

Scaling: Partition by title/room ID, sticky sessions, shard warmup, tree fanout for large rooms.

Algorithms: Ordered delivery, transactional outbox, hybrid fanout, TTL heartbeats, inline + ML moderation, graceful degradation.

Security: TLS 1.3, KMS encryption, pseudonymous IDs, RBAC/ABAC, multi-AZ, regional residency, WAF & bot protection.

Operations: Multi-tenancy with quotas, progressive delivery, observability (Prometheus, OpenTelemetry), SLO-aware autoscaling, chaos drills, DR.

Capacity & Cost: Busy region: ~1M users, 10K–50K msgs/sec. Infra: Kafka, Redis (12 shards), DynamoDB/Scylla.
Monthly cost ~\$30K–\$80K/region.

Build Plan: MVP (3 mo): basic chat, single region. Phase 2 (6–9 mo): multi-region, presence, typing,

analytics.

Phase 3 (12 mo): ML moderation, tiered fanout, premium SLOs.

Risks: Kafka ops overhead, Redis memory, celebrity room hotspots, compliance fragmentation.